Mario Torres Danta

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Seville, Spain

Education

B.Sc. in Mathematics *2023 – 2027 (Expected)*

University of Seville (US), Seville, Spain

Current GPA: 8.74 / 10.0 (Top 5% of cohort; based on 22 graded courses)

Bachelor's thesis (in progress): Algebraic K-theory. Supervisor: Prof. Ramón J. Flores.

Academic Honors and Distinctions

Collaboration Grant (*Beca de Colaboración*), Spanish Government *2026 – 2027*

Competitive national research-initiation grant for top final-year students, held in a university department.

Awarded **"Matrícula de Honor" (Highest Distinction)** in several courses, such as:

- **General Topology (10.0)**
- **Mathematical Logic (10.0)**
- **Measure Theory & Integration (10.0)**
- **Mathematical Optimization (9.2)**
- **Numerical Analysis I (9.5)**
- **Ordinary Differential Equations (9.4)**
- **Probability Theory (9.4)**

Additional "Sobresaliente" (A-Grade) in courses such as Local Geometry of Surfaces (10.0), Algebraic Structures, Complex Analysis, Linear Algebra II, Discrete Mathematics.

Research Experience

Visiting Student & Undergraduate Researcher *Aug 2026 – Sep 2026*

University of Copenhagen, Denmark

Project: Mapping class groups of finite abelian branched covers of surfaces. Identified the liftable and symmetric mapping class groups with mixed braid groups, described the lifts explicitly via chain twists and multi-Dehn twists, and derived homological stability results. Resulted in a manuscript in preparation (see below).

Supervisor: Prof. Nathalie Wahl

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FUSRP Undergraduate Researcher *Jun 2026 – Jul 2026*

The Fields Institute for Research in Mathematical Sciences, Toronto, Canada

Project: Laurent symmetric functions, which extend classical symmetric functions to the rational representations of general linear groups. Studied their ring structure, constructed orthonormal bases for the Hall inner product, and applied them to representations of GL_n and GL_∞. Resulted in a preprint (see below).

Supervisors: Prof. Alistair Savage and Dr. Yaolong Shen

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Undergraduate Researcher (Departmental Collaborator) *Sep 2025 – Jun 2026*

Department of Geometry and Topology, University of Seville

Project: Advanced topics in Topology and Group Theory. A guided study complementing the standard undergraduate curriculum.

Supervisor: Prof. Ramón J. Flores

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Undergraduate Researcher (Departmental Collaborator) *Sep 2024 – Jun 2025*

Department of Algebra, University of Seville

Project: Classification of homographies and homologies in real ($\mathbb{P}^n(\mathbb{R})$) and complex ($\mathbb{P}^n(\mathbb{C})$) projective spaces.

Supervisor: Prof. Francisco Castro Jiménez

Publications and Manuscripts

H. Fernandez, E. Kleinwort, T. Li and **M. Torres-Danta**, *Laurent symmetric functions*, preprint (2026). arXiv:2609.15690.

M. Torres-Danta, *The symmetric mapping class group of finite abelian branched covers*, manuscript in preparation (2026).

Talks

Laurent symmetric functions *Jul 2026*

Final presentation, Fields Undergraduate Summer Research Program, Fields Institute, Toronto.

Mapping class groups of finite abelian branched covers *Sep 2026*

Department of Mathematical Sciences, University of Copenhagen.

Research Interests

Algebraic Topology (Homology, Homotopy Theory, Homological Stability) · Geometric Topology (Mapping Class Groups, Moduli Spaces) · Algebraic K-Theory · Group Theory · Category Theory

Academic Service

Reviewer, zbMATH Open *Sep 2026 – Present*

Writing reviews of recent research articles for the zbMATH Open database.

Academic Engagement

ByMat Conference (Bringing Young Mathematicians Together) *Nov 2025*

University of Seville / RSME (Royal Spanish Mathematical Society)

Role: Selected attendee. Highlights: plenary lectures by **Prof. Marithania Silvero** (Knot Theory) and **Prof. Alberto Rodríguez** (Differential Geometry/Topology).

Skills

- **Languages:** Spanish (Native), English (C1), French (B2)
- **Computing: Python** (*Santander Open Academy* certificate), **MATLAB**, **LaTeX**, **GitHub**

References

Prof. Nathalie Wahl
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Web: web.math.ku.dk/~wahl

Prof. Alistair Savage
Professor, Dept. of Mathematics and Statistics, University of Ottawa
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Prof. Ramón J. Flores
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